Shobhan Roy, PhD

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SUMMARY

Computational physicist and SciML researcher designing hybrid physics + ML pipelines for advanced materials research.

- Leading the design of an agentic microstructure-to-performance sub-pipeline that routes real (SEM, µ-CT) and ML-generated micrographs through multi-modal pathways — either physics-based (**SCIMITAR3D**) or neural (**PARC-family**) predictors — handling data/memory management and cross-platform interfacing; part of a broader materials-discovery framework.
- Co-architected **D-PARC**, a deformable-convolution neural operator for nonlinear transport (Beerman, Roy, et al., *arXiv* 2026); simulation-side technical lead across multi-institution SciML programs, defining ground-truth DNS datasets and physics-consistency criteria — conservation, interface, entropy — for model predictions.
- **7 peer-reviewed journals** (*Shock Waves*, *J. Fluid Mechanics*, *J. Applied Physics*, *APL Machine Learning*); production HPC at **150M-cell, 7,000-core, ~3M CPU-hour** scale on DoD clusters.

CORE SKILLS

- **Scientific ML:** physics-aware neural operators (PARC, D-PARC), surrogate modeling, physics-consistent evaluation (conservation / interface / entropy checks), uncertainty quantification, pipelines integrating ML-generated microstructures (diffusion models, U-Net, CycleGAN) into physics solvers; *adjacent / conversant:* PINNs, FNO, DeepONet, differentiable simulation, active learning, Bayesian optimization.
- **Scientific Computing & HPC:** Python, Fortran 90, C++, Bash; MPI, domain decomposition, Slurm/PBS, parallel I/O; multi-material flow-solver components (level-set, ghost-fluid method); finite-volume methods and shock-capturing schemes; multi-thousand-core campaigns on DoD HPC clusters.
- **Frameworks, MLOps-adjacent practice & tooling:** PyTorch, TensorFlow; Git, LaTeX, Jupyter; model registry, provenance / lineage tracking, RAG over scientific data; multi-agent orchestration; LLM-assisted research workflows (VS Code agent mode, Github Copilot CLI) for code review, scheme prototyping, and reproducible build pipelines.

EXPERIENCE

Postdoctoral Research Scholar — IIHR–Hydroscience & Engineering, University of Iowa

May 2025 – Present · Iowa City, IA

- Co-architected **D-PARC** — a physics-aware deformable-convolution neural operator built on the **PARC** family for emulation of high-speed nonlinear transport — contributing conceptual analogies between hybrid Lagrangian–Eulerian flow-solver schemes and the deformable-convolution operator (Beerman, Roy, et al., *arXiv* 2026).
- Co-designing (with PI) an agentic, closed-loop materials-discovery framework integrating generative inverse design; leading the design of the framework's microstructure-to-prediction sub-pipeline — routing real (SEM, µ-CT) and ML-generated micrographs into either physics-based (**SCIMITAR3D**) or neural (**PARC-family**) predictors, making the architectural calls on where to inject agentic AI, how to manage data and provenance across simulation and experimental stores, and how to interface heterogeneous compute and storage; mentoring **1 postdoc, 2 graduate students, and 1 undergraduate intern**.
- Designed and executed high-resolution simulations of shock-mediated damage collapse and hotspot formation in energetic materials to benchmark physics-aware deep-learning surrogates against physically critical quantities (peak pressure, temperature localization, reaction-front velocity).
- Architecting novel interfacial treatments for materials undergoing solid → gaseous (hydrodynamic) phase change, extending the multi-material solver to seamlessly handle chemically decomposing systems; extended **Mie–Grüneisen** and **JWL EOSs** to handle tensile-to-high-compression pressure range required for non-planar flow in 2D/3D geometries.
- Interfaced with **University of Virginia's ML group (Baek lab)** as the simulation-side lead — curating ground-truth DNS databases, selecting simulation parameters to maximize physical coverage, defining validation benchmarks for training **PARC**.
- *External technical engagements:* simulation-side reference for junior researchers at **IIT Delhi** and **IIT Madras** on a co-developed multi-physics solver; contributing to a newly awarded **ONR-MURI** on non-equilibrium reactive dynamics in energetic materials.

Graduate Research Assistant (PhD) — University of Iowa

Aug 2019 – Apr 2025 · Iowa City, IA

- Extended **SCIMITAR3D** (in-house continuum-mechanics hydrocode) into a multi-material sharp-interface framework for shock-to-detonation transition in heterogeneous energetic materials; integrated 5th-order WENO reconstruction and Riemann-based ghost-fluid coupling — eliminating the **3× grid-refinement overhead** required for equivalent accuracy at lower order.
- Designed and executed production-scale HPC campaigns on DoD clusters — **150M grid cells, 7,000 CPU cores, ~3M CPU-hours** — at **15 nm grid resolution** on meso-scale sub-samples (50 × 70 µm, 1–2 energetic crystals) (*Shock Waves*, 2026).
- Validated mesoscale shock-initiation predictions against experimental data from collaborators at **UIUC (Dlott lab)** and **LANL**, selecting constitutive models and resolution criteria for head-to-head comparison at sub-grain fidelity. Bridged simulation contributions across a **7-university AFOSR-MURI** (*Microstructurally-Aware Continuum Models for Energetic Materials*, 2019–2024) consortium.
- Led simulation and physics-validation for **HEDS** *(Heterogeneous Energetic Material Damage Simulator)* — a deep-learning + computational-mechanics pipeline for microstructure-to-sensitivity mappings; contributed evaluation criteria, simulation integration, and interpretation of learned representations (*APL Machine Learning*, 2025).

Graduate Research Scholar (MS) — IIT Madras, India

Jul 2015 – Jun 2019 · Chennai, India

- Characterized normal-shock / boundary-layer interaction (SBLI) and porous-medium flow control through combined CFD simulations and transonic wind-tunnel experiments; designed finite-volume RANS campaigns to evaluate shock-control effectiveness, validated against experimental data (*Shock Waves* 2021; *AIAA Journal* 2019).
- Formulated an analytical model for asymmetric wedge-induced Mach reflection in 2D steady supersonic flows (parallel research; *Journal of Fluid Mechanics* 2019). Subsequently contributed to **DRDO's Variable Camber Morphing Wing** project (2018–2019) at IIT Madras's Center for Propulsion Technologies through immersed-boundary implementation and preliminary aerodynamic analysis.

SELECTED PUBLICATIONS

Full list of peer-reviewed journals (7), conference proceedings (4), and presentations (14) at [Google Scholar]() and [personal site]().

1. Beerman, J.T., **Roy, S.**, Udaykumar, H.S., Baek, S.S. — *Size is Not the Solution: Deformable Convolutions for Effective Physics-Aware Deep Learning*, **arXiv preprint** (2026).
2. **Roy, S.**, Seshadri, P.K., Okafor, C., Johnson, B.P., & Udaykumar, H.S. — *High-Fidelity Simulations of Shock Initiation of an Energetic Crystal–Binder System Due to Flyer Impact*, **Shock Waves** 36(2), 2026.
3. Fang, I., **Roy, S.**, Nguyen, P.C.H., Baek, S.S., Udaykumar, H.S. — *Heterogeneous Energetic Material Damage Simulator (HEDS): A Deep Learning Approach to Simulate Damage–Sensitivity Linkages*, **APL Machine Learning** 3(2), 2025.

AWARDS & SERVICE

- ***Peer Reviewer****: Journal of Applied Physics* (AIP) · *Physics of Fluids* (AIP) · *AIAA Journal*
- ***Invited Judge,*** Jakobsen Graduate Research Showcase, University of Iowa Graduate College (2026)
- **Ballard & Seashore Dissertation Fellowship**, University of Iowa (2025)
- **Finalist (Top 13)**, University of Iowa Graduate College Three Minute Thesis (3MT), selected through a graduate-college-wide preliminary round (2024)
- **Excellent Presentation Award**, 7th International Symposium on Energetic Materials and Their Applications, Tokyo, Japan (2021)
- **Dean's Graduate Fellowship**, College of Engineering, University of Iowa (2019)

EDUCATION

PhD, Mechanical Engineering — University of Iowa, 2025 · Advisor: Prof. H.S. Udaykumar
Dissertation: A versatile framework for high-fidelity computations of high-speed reactive multi-material dynamics.

MS (Research), Aerospace Engineering — IIT Madras, India, 2019 · Advisors: Prof. Santanu Ghosh & Prof. G. Rajesh
Thesis: Investigation of drag reduction using a porous medium for a normal-shock/boundary-layer interaction.

BTech, Mechanical Engineering — NIT Durgapur, India, 2015